SRM Institute of Science and Technology
Department of Mathematics
21MAB204T-Probability and Queueing Theory
Unit – I
Tutorial Sheet - 2

S.No.	Part – B (Each question carries 8 marks)
1	When a die is thrown, X denotes the number that turns up. Find $E(X)$, $E(X^2)$, $\text{Var}(X)$, and Standard Deviation (SD).
2	The Probability Distribution Function of a RV X is given below: x: 0, 1, 2, 3, 4 $p(x)$: 1/16, 4/16, 6/16, 4/16, 1/16 Find the MGF of X and hence find (i) the mean of X (ii) the variance of X (iii) μ_3'
3	Find the Moment Generating Function of the continuous RV X whose density function is $f(x) = \begin{cases} 2e^{-2x} & , x \geq 0 \\ 0 & , otherwise \end{cases}$. Hence find the first four central moments.
4	The first three moments of a distribution about $X = 4$ are 1, 4, 10 respectively. Find the first three moments about the origin and also find the Standard Deviation (SD).
5	The Random Variable X has the following probability distribution: x: 0, 1, 2, 3, 4 $p(x)$: K, $3K$, $5K$, $7K$, $9K$ Find (i) K, (ii) mean of X (iii) variance of X (iv) $E(3X-4)$ and (v) $\text{Var}(3X-4)$
	Part – C (Each question carries 15 marks)
6	The Probability Distribution Function of a discrete RV X is given by x: 0, 1, 2, 3 $p(x)$: k, $3k$, $5k$, $7k$ Find (i) k, (ii) CDF (iii) $E(X)$ (iv) $V(X)$ (v) $E(4X+5)$ (vi) $V(4X+3)$ and (vii) MGF of X
7	A continuous RV X has pdf $f(x) = \begin{cases} k(1-x) & , 0 < x < 1 \\ 0 & , otherwise \end{cases}$ Find (i) k (ii) the rth moment about the origin. Hence find the first four central moments.
8	If the p.d.f $f(x) = \begin{cases} x & , 0 \leq x \leq 1 \\ 2-x & , 1 \leq x \leq 2 \\ 0 & , otherwise \end{cases}$ Compute the rth moment about the origin. Hence deduce the first four moments about mean.